

Does Test-Time Scaling via Budget Forcing Transfer to Vietnamese Reasoning?

Van Hong Hanh Phan Hong Hanh
University of Technology, Ho Chi Minh City, Vietnam

Abstract

Test-time scaling improves language-model reasoning by allocating additional computation during inference rather than only during training. This report reviews Muenighoff et al. (2025), which combines a compact reasoning dataset with budget forcing, and then studies whether the budget-forcing component transfers to Vietnamese-language reasoning. We adapt the decoding trigger to Vietnamese, sweep $n_{\text{wait}} \in \{0, 1, 2\}$, and evaluate reasoning-distilled, Vietnamese-specialized, and Southeast Asian open models on Vietnamese math, factual multiple-choice, and exploratory high-school exam benchmarks. The study separates mechanical control over thinking length from useful accuracy gains, asking whether extra generated reasoning actually improves answers under realistic compute constraints.

1 Introduction

Large language model progress has traditionally been explained through train-time scaling: larger models, larger corpora, and longer optimization runs. Reasoning-oriented systems have added a second axis, *test-time scaling*, where extra computation is spent during generation itself. Instead of changing model parameters, the system can ask the model to produce longer intermediate reasoning, revisit an earlier step, sample multiple candidate answers, or search over possible solutions. This shift is important for researchers with limited training compute because it makes inference-time behavior an object of direct experimental control.

The s1 paper is a useful reference point because it asks how simple an open test-time scaling recipe can be (Muenighoff et al., 2025). Its recipe has two parts. First, the authors curate **s1K**, a compact set of 1,000 high-quality reasoning examples distilled from a much larger candidate pool. Second, they introduce **budget forcing**, a decoding-time intervention that controls how long a model thinks. If the model tries to end its reasoning too early, the decoder suppresses the end-of-thinking delimiter and appends a continuation phrase such as “Wait”. If the model exceeds a maximum budget, the decoder forces the transition to the final answer. This mechanism is attractive because it does not require reinforcement learning, new model weights, or access to proprietary training pipelines.

However, the central empirical question is whether budget forcing is a general reasoning primitive or a method

whose gains depend on the original language, model family, and benchmark mix. The strongest s1 results are reported on English reasoning benchmarks such as AIME24, MATH500, and GPQA Diamond with a Qwen2.5-based reasoning model. Vietnamese introduces a different transfer setting: the prompt language changes, answer formats vary across math and multiple-choice tasks, and several candidate models are instruction-tuned for Vietnamese use rather than explicitly trained to emit long self-reflective traces. A method that increases thinking tokens may therefore show good *control* while still failing to improve accuracy.

This report studies that transfer problem. We focus on budget forcing rather than reproducing the full s1 supervised fine-tuning pipeline. The English trigger is replaced with the Vietnamese phrase “*Chờ một chút*”, model-specific end-of-turn or end-of-thinking delimiters are registered, and the evaluation sweeps $n_{\text{wait}} \in \{0, 1, 2\}$ with $n_{\text{wait}}=0$ as the greedy baseline. The model set covers a Vietnamese reasoning-oriented baseline, a reasoning-distilled Qwen model inspired by the s1 backbone, and Vietnamese or Southeast Asian chat models. The benchmark set covers Vietnamese mathematical reasoning, factual multiple-choice knowledge, and an exploratory Vietnamese high-school exam setting when result artifacts are available.

The study is guided by four research questions. First, does budget forcing produce a positive scaling signal on Vietnamese math? Second, does the signal weaken or reverse on factual and exam-style multiple-choice tasks where extra reasoning may not add missing knowledge? Third, do Vietnamese-specialized chat models respond differently from reasoning-distilled models? Fourth, can the method reliably increase thinking length without causing loops, delimiter errors, or answer-extraction failures?

In summary, this report contributes: (i) a concise reconstruction of the s1 test-time scaling method; (ii) a Vietnamese adaptation of budget forcing using a native continuation trigger and model-family delimiter handling; (iii) a cross-model evaluation protocol over greenmind-14b-r1, r1-distill-7B, vinallama-7b, vistrail-7b, and seallm-7b; and (iv) an analysis framework that separates mechanical increases in thinking tokens from downstream accuracy gains. Section 2 describes the method, Section 3 defines the metrics, Section 4 analyzes expected limitations, Section 5 situates the work, Section 6 reports experiments, and Section 7 concludes.

2 Method

2.1 s1 as the Reference Method

The s1 framework combines small-scale reasoning-data curation with an inference-time control mechanism (Muenighoff et al., 2025). Its data component, **s1K**, is selected from roughly 59K candidate problems using three filters: quality, difficulty, and diversity. Quality filtering removes noisy or poorly formatted traces; difficulty filtering emphasizes problems where additional reasoning is useful; and diversity filtering prevents the final 1K set from collapsing into repeated topics. The selected questions are paired with long reasoning traces distilled from a stronger thinking model, then used for supervised fine-tuning of a Qwen2.5-32B-Instruct base model.

Our study does not reproduce this supervised fine-tuning stage. Instead, we isolate the second part of s1: budget forcing at decoding time. This choice makes the experiment feasible under limited hardware and directly tests the most portable part of the paper. The resulting study should therefore be read as a transfer evaluation of budget forcing, not as a complete reproduction of s1-32B.

2.2 Budget Forcing

Budget forcing controls the length of the model’s reasoning trace by intervening when special delimiters are produced. Let n_wait denote the number of times the decoder is allowed to prevent early stopping. When the model emits an end-of-thinking or end-of-turn token before the configured minimum has been reached, the decoder suppresses that token and appends a short continuation trigger. In s1, the trigger is “Wait”. The appended phrase pushes the model to continue the autoregressive trace, often causing verification, alternative derivations, or correction of an earlier answer.

The same mechanism can also enforce an upper budget. If the model continues past the maximum allowed thinking length, the decoder appends the final-answer prefix or the appropriate delimiter so that generation exits the reasoning region. The method is simple but not risk-free: extra tokens can become repetition, a suppressed delimiter can break the chat template, and longer traces can make answer extraction harder. For this reason, our evaluation reports both token-length control and answer accuracy.

2.3 Vietnamese Adaptation

The Vietnamese transfer setting changes both the language of the continuation trigger and the token conventions used by different model families. We replace “Wait” with “*Chờ một chút*”, a natural Vietnamese pause phrase that encourages reconsideration without switching back to English. We also maintain a delimiter registry for common chat and reasoning formats, including `<|im_end|>` for Qwen-style models, `</think>` for reasoning models, and `</s>` for LLaMA- or Mistral-style chat models.

All experiments sweep $n_wait \in \{0, 1, 2\}$. The setting $n_wait=0$ is the greedy baseline: the model is allowed

Table 1: Model families used in the Vietnamese budget-forcing study.

Model	Alignment role	Reasoning role
greenmind-14b-r1	Vietnamese	Reasoning-oriented baseline
r1-distill-7B	English-primary	Reasoning-distilled
vinallama-7b	Vietnamese	Chat model
vistral-7b	Vietnamese	Chat model
seallm-7b	Southeast Asian	Chat model

Table 2: Benchmark roles and answer formats.

Benchmark	Task type	Answer
vi_gsm8k	Math reasoning	Number
vimmlu	Factual MCQ	A/B/C/D
vnhsge	Exam MCQ	Normalized choice

to stop normally. Settings $n_wait=1$ and $n_wait=2$ test whether one or two forced continuations increase useful reasoning. Because longer traces increase decoding cost, the implementation relies on cached key-value states rather than recomputing the full prefix after every intervention.

2.4 Models

The model set is designed to separate language alignment from reasoning alignment. `greenmind-14b-r1` is the strongest Vietnamese reasoning-oriented checkpoint in the available artifacts and provides a high baseline for judging whether forcing is still useful when the model already answers well. `r1-distill-7B` represents a reasoning-distilled model built on a Qwen backbone and is expected to be more compatible with long deliberative traces (DeepSeek-AI et al., 2025; Qwen Team, 2024). `vinallama-7b`, `vistral-7b`, and `seallm-7b` represent Vietnamese-specialized or Southeast Asian chat models that may understand Vietnamese better but were not necessarily post-trained for extended self-reflective reasoning.

2.5 Benchmarks and Answer Extraction

We evaluate on Vietnamese tasks with different expected relationships between reasoning length and correctness. `vi_gsm8k` measures multi-step arithmetic reasoning and uses numeric answer extraction. `vimmlu` measures factual and multi-disciplinary multiple-choice knowledge, where extra reasoning may not help if the missing information is not already in the model. `vnhsge` is treated as an exploratory Vietnamese high-school exam benchmark when result artifacts are present; its answers are normalized from indexed choices to a common multiple-choice format.

The evaluation driver records the generated answer, extracted answer, correctness, n_wait , and thinking-token count for each sample. Aggregation then produces accuracy, scaling slope over n_wait , maximum performance

across the sweep, average thinking tokens, and extraction failures. This design follows the s1 distinction between *control*, where a method successfully changes the amount of test-time computation, and *scaling*, where the extra computation improves task accuracy.

3 Evaluation Metrics

The s1 paper evaluates test-time scaling with three complementary ideas: control, scaling, and performance (Muenighoff et al., 2025). We keep that structure but add two implementation-facing measures needed for Vietnamese answer extraction.

3.1 Control

Control measures whether the intervention actually changes the amount of test-time computation. In this report, the most direct proxy is the average number of thinking tokens generated at each `n_wait` value. If `n_wait` increases but token counts remain flat, the decoder is not reliably controlling the budget. If token counts increase while accuracy does not, budget forcing has mechanical control but weak useful transfer.

3.2 Scaling

Scaling measures how accuracy changes as more inference-time compute is allocated. The repository estimates this with the slope of accuracy over the `n_wait` sweep. A positive slope indicates that longer controlled reasoning is associated with better answers; a negative slope indicates that forced continuation is counterproductive for the model or benchmark.

3.3 Performance

Performance is the maximum accuracy achieved across tested budgets. It is useful for comparing the strongest observed setting, but it should not be interpreted alone. A model can have high peak accuracy with flat scaling, or it can increase tokens substantially while losing accuracy.

3.4 Extraction Reliability

Vietnamese evaluation also depends on robust answer parsing. We therefore report extraction failures, defined as cases where the parser cannot identify a normalized numeric answer or multiple-choice option. This metric helps distinguish reasoning failures from formatting failures caused by loops, delimiter mismatch, or prompt contamination.

4 Strengths, Limitations, and Research Gaps

4.1 Strengths

Budget forcing is attractive because it is transparent and locally inspectable. Unlike reinforcement-learning-heavy rea-

soning pipelines, the core intervention is a decoding wrapper: suppress an early delimiter, append a trigger, and continue generation. This makes the method suitable for a coursework reproduction and for cross-lingual stress tests where retraining a large model is unrealistic.

A second strength is decomposition. Even without reproducing the full s1 training recipe, the report can test whether its most portable inference-time component behaves as expected on new models and tasks. This directly supports the distinction between a model that can be controlled to think longer and a model that benefits from thinking longer.

4.2 Limitations

The method depends on model-specific chat templates and delimiter conventions. A Qwen-style end token, a reasoning-model `</think>` token, and a LLaMA-style `</s>` token do not have identical semantics. Suppressing the wrong token can break the conversation format or make the model continue in an unnatural mode.

Budget forcing also assumes that a fixed trigger phrase elicits useful reflection. This is not guaranteed. Forced continuation may correct an arithmetic mistake, but it may also produce repetitive text, overthink a correct answer, or drift away from the required answer format. These risks are especially important for multiple-choice benchmarks, where concise factual recall can outperform long deliberation.

4.3 Research Gaps

The main gap addressed here is Vietnamese transfer. The original s1 evidence is strongest for English reasoning benchmarks and a Qwen2.5-based reasoning model. This report asks whether the same decoding control signal transfers to Vietnamese prompts, Vietnamese-specialized chat models, and benchmarks that mix mathematical reasoning with factual or exam-style multiple-choice questions.

A second gap is trigger design. The phrase “*Chờ một chút*” is a natural first adaptation, but future work should compare variants such as “*Đợi đã*” or “*Suy nghĩ thêm*” and should add anti-loop safeguards. Finally, there remains a broader contrast between internal-knowledge scaling and retrieval-augmented methods: budget forcing tries to improve how a model uses its existing knowledge, while retrieval improves what evidence is available at inference time (Li et al., 2025).

5 Related Work

5.1 Reasoning-oriented test-time scaling

Budget forcing belongs to a broader family of methods that attempt to improve reasoning by increasing or restructuring inference-time computation. The recent wave of reasoning-focused systems includes large proprietary models and open reproductions, but many of them rely on training pipelines or infrastructure that are difficult to reproduce directly. DeepSeek-R1 demonstrates that reinforcement learning can induce self-reflective reasoning behav-

iors at scale (DeepSeek-AI et al., 2025). Kimi k1.5 similarly explores stronger reasoning through large-scale reinforcement learning and long-context optimization (Kimi Team et al., 2025). Against this backdrop, s1 is notable because it argues for a minimal recipe: a compact curated dataset and a decoding intervention rather than a heavy post-training stack (Muennighoff et al., 2025). Recent systems such as Sky-T1 also pursue open reasoning recipes that reduce dependence on closed-source pipelines (Zhang et al., 2025).

The repository used in this project aligns with that minimal philosophy. It does not attempt to compete with large closed systems directly. Instead, it asks whether the same qualitative trend, namely better answers under controlled increases in test-time compute, can be observed in a small and reproducible setup.

5.2 Model backbone and evaluation tooling

The experimental path in this repository relies on open-weight reasoning or instruction-tuned backbones, particularly DeepSeek-R1 distilled checkpoints and Qwen2.5 variants. Qwen2.5 is a relevant backbone family because it provides instruction-tuned models across multiple sizes and explicitly emphasizes long-context and structured-output capabilities (Qwen Team, 2024). For evaluation tooling and reproducibility practices, the project also draws on the design philosophy of the Language Model Evaluation Harness, which standardizes benchmark execution and result logging across model backends (Gao et al., 2024).

5.3 Inference-time augmentation beyond budget forcing

Inference-time improvement is not limited to longer reasoning traces. Retrieval-augmented generation, for example, supplements the model’s internal knowledge with external evidence rather than extending its internal chain of thought. The systematic survey by Li et al. (2025) shows how retrieval augmentation can improve reliability, freshness, and domain grounding in educational applications. This contrast is useful for the present report. Budget forcing attempts to extract more value from a model’s internal reasoning process, whereas retrieval augmentation changes the information available at inference time. Both approaches increase inference-time effort without retraining the base model, but they solve different failure modes.

For that reason, the comparison between budget forcing and retrieval-augmented generation should be treated as complementary rather than competitive. One extends internal deliberation; the other injects external evidence. A practical future direction is to study whether the two can be combined, for example by retrieving task-relevant context and then allocating a controllable reasoning budget over that context.

Table 3: Accuracy change from $n_wait=0$ to $n_wait=2$ in percentage points.

Model	vi_gsm8k	vimmlu	vnhsge
r1-distill-7B	+4	+4	+3
seallm-7b	-19	+10	+6
vinallama-7b	-9	-13	-5
vistral-7b	-5	-26	-33
greenmind-14b-r1	-25	-10	0

6 Experiments

6.1 Run Protocol

The Vietnamese evaluation uses a budget-forcing-only design. Retrieval augmentation and new supervised fine-tuning are out of scope. For each model–benchmark pair, the driver evaluates $n_wait \in \{0, 1, 2\}$, records generated outputs and extracted answers, and aggregates accuracy, scaling slope, maximum performance, average thinking tokens, and extraction failures.

The available matrix contains five models and three Vietnamese benchmarks with 100 examples per model–benchmark–budget setting. `vi_gsm8k` tests mathematical reasoning with numeric answer extraction, `vimmlu` tests factual multiple-choice knowledge, and `vnhsge` adds an exam-style multiple-choice stress test. The result artifacts are summarized in `figures/summary_vi.csv` and visualized in Figures 1–3.

6.2 Accuracy Under Budget Forcing

Figure 1 shows that Vietnamese transfer is uneven rather than uniformly positive. On `vi_gsm8k`, `greenmind-14b-r1` starts from the strongest baseline at 80% accuracy but drops to 55% at $n_wait=2$. `seallm-7b` also declines sharply on the math task, from 66% to 47%. The clearest positive math result is `r1-distill-7B`, which improves from 53% to 57%, suggesting that a reasoning-distilled Qwen-family model is more compatible with forced continuation than the Vietnamese chat models in this setting.

The multiple-choice benchmarks show a different pattern. On `vimmlu`, `seallm-7b` rises from 46% to 56%, while `r1-distill-7B` rises from 27% to 31%. However, `vistral-7b` falls from 39% to 13%, and `vinallama-7b` falls from 34% to 21%. On `vnhsge`, `seallm-7b` again improves, from 47% to 53%, `r1-distill-7B` improves from 27% to 30%, and `greenmind-14b-r1` remains tied at 57% for $n_wait=0$ and $n_wait=2$. These results indicate that budget forcing is not simply “more reasoning is better”; it is a model-specific perturbation whose benefits depend on whether the continuation resembles useful verification.

6.3 Token-Budget Control

Figure 2 confirms that the decoder usually changes the amount of generated reasoning. For `r1-distill-7B`,

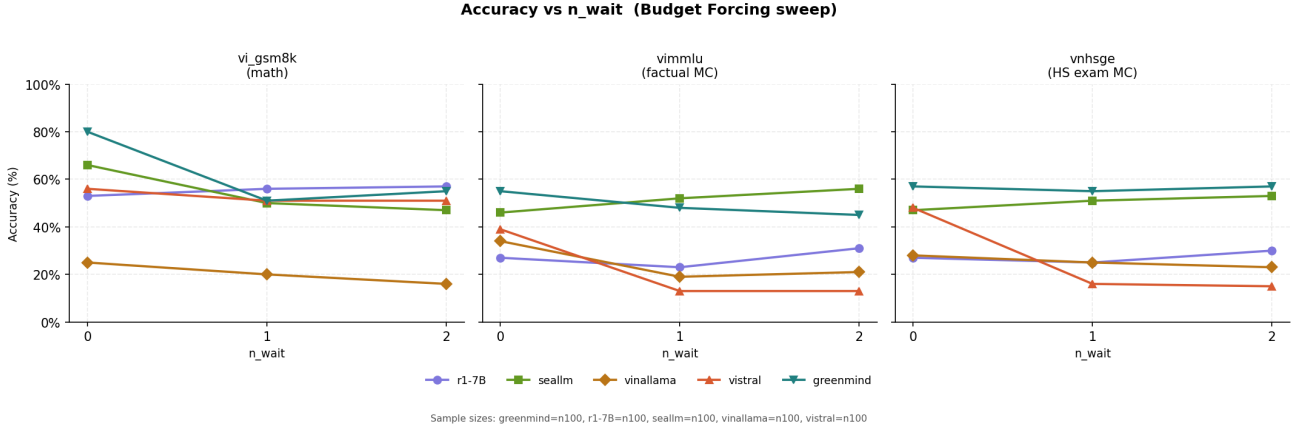


Figure 1: Accuracy across the n_wait sweep. Budget forcing helps only for selected model–benchmark pairs; several strong baselines lose accuracy after forced continuation.

average thinking tokens increase from 191.0 to 426.1 on *vi_gsm8k*, from 539.8 to 870.6 on *vimmlu*, and from 568.1 to 848.1 on *vnhsge*. *seallm*-7b shows even stronger token growth on *vi_gsm8k*, from 236.2 to 734.4 tokens, but its accuracy drops by 19 percentage points on that benchmark. This is the clearest example of mechanical control without beneficial scaling.

The main anomaly is *greenmind*-14b-r1 on *vi_gsm8k*, where the average thinking length decreases from 199.4 tokens at $n_wait=0$ to 165.3 at $n_wait=2$. This suggests that the delimiter or continuation intervention may not be aligned with that model’s generation format on the math task. In contrast, *vistrail*-7b on *vnhsge* expands dramatically, from 50.5 to 906.8 tokens, while accuracy collapses from 48% to 15%. The failure mode is therefore not insufficient budget; it is low-quality or format-disruptive continuation.

6.4 Net Budget-Forcing Effect

Figure 3 and Table 3 summarize the net effect of moving from the greedy baseline to two forced continuations. Only *r1-distill*-7B improves on all three benchmarks, with gains of +4, +4, and +3 percentage points. *seallm*-7b is mixed: it improves on both multiple-choice tasks but loses 19 points on math. *vinallama*-7b, *vistrail*-7b, and *greenmind*-14b-r1 mostly decline, despite often producing longer traces.

Overall, the Vietnamese results support a conservative interpretation of budget forcing. The method reliably exposes whether a model can be pushed into longer reasoning, but longer reasoning is useful only when the model has learned a compatible verification behavior. In these artifacts, *r1-distill*-7B provides the most consistent positive transfer, while several Vietnamese-oriented chat models demonstrate that native-language competence alone is not enough for beneficial test-time scaling.

Table 4: Best observed accuracy across the budget sweep.

Benchmark	Best model and setting	Accuracy
<i>vi_gsm8k</i>	<i>greenmind</i> -14b-r1, $n_wait=0$	80%
<i>vimmlu</i>	<i>seallm</i> -7b, $n_wait=2$	56%
<i>vnhsge</i>	<i>greenmind</i> -14b-r1, $n_wait=0/2$	57%

6.5 Extraction Reliability

All reported rows have zero extraction failures. This strengthens the interpretation that the observed drops are primarily reasoning or generation-quality failures rather than parser failures. Nevertheless, the large token increases for *vistrail*-7b and *vinallama*-7b suggest that future error analysis should inspect individual traces for repetition, answer drift, and delimiter mismatch. Aggregate curves identify where budget forcing helps or hurts; qualitative examples are still needed to explain why a forced continuation corrects one model and destabilizes another.

7 Discussion and Conclusion

This report reviews *s1* as a simple open recipe for test-time scaling and isolates budget forcing as the component most suitable for Vietnamese transfer. The central lesson from the Vietnamese artifacts is that inference-time compute is often controllable, but controllability alone is not equivalent to reasoning improvement. The figures show that many models generate longer traces under $n_wait=2$, yet only *r1-distill*-7B improves consistently across all three benchmarks. *seallm*-7b benefits on multiple-choice tasks but loses accuracy on math, while *vistrail*-7b, *vinallama*-7b, and *greenmind*-14b-r1 mostly show negative or flat budget-forcing deltas.

The Vietnamese study is intentionally narrow: no new *s1K*-style Vietnamese fine-tuning, no retrieval augmentation, and a small n_wait sweep. Within that scope, the method provides a clear way to compare reasoning-distilled

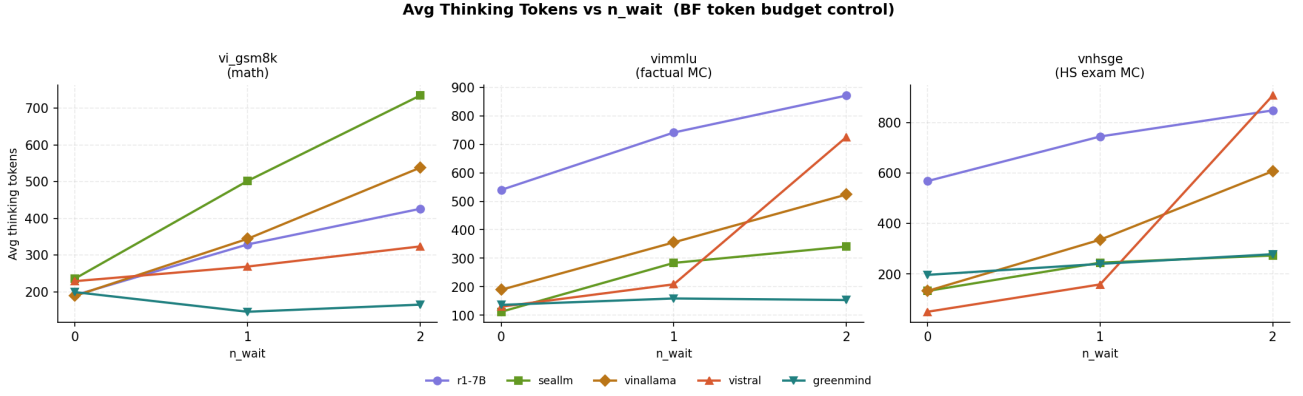


Figure 2: Average thinking-token counts across the n_{wait} sweep. Most models generate longer traces when forced, but longer traces do not consistently translate into higher accuracy.

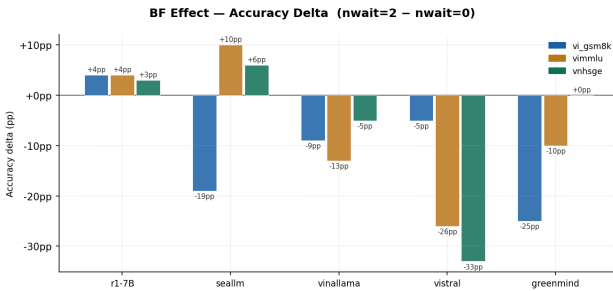


Figure 3: Accuracy delta from $n_{wait}=0$ to $n_{wait}=2$. Positive transfer is concentrated in r1-distill-7B and selected seallm-7b multiple-choice settings.

and Vietnamese-specialized models under the same decoding intervention. The most important interpretation is therefore conditional: budget forcing is useful when extra thinking tokens become verification or correction, and it is harmful when they amplify loops, overthinking, or answer drift.

Future work should build a Vietnamese reasoning dataset analogous to s1K, compare multiple Vietnamese trigger phrases, add anti-loop stopping rules, and evaluate harder Vietnamese math and science benchmarks. These extensions would test whether the positive scaling behavior reported by s1 is a universal decoding phenomenon or a property that emerges only when the model, data, language, and task are aligned.

References

DeepSeek-AI, Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Peiyi Wang, Qihao Zhu, Runxin Xu, and Ruoyu Zhang. Deepseek-r1: Incentivizing reasoning capability in LLMs via reinforcement learning. *arXiv*

preprint *arXiv:2501.12948*, 2025. URL <https://arxiv.org/abs/2501.12948>.

Leo Gao, Jonathan Tow, Baber Abbasi, Stella Biderman, Sid Black, Anthony DiPofi, Charles Foster, Laurence Golding, Jeffrey Hsu, Alain Le Noac’h, Haonan Li, Kyle McDonell, Niklas Muennighoff, Chris Ociepa, Jason Phang, Laria Reynolds, Hailey Schoelkopf, Aviya Skowron, Lintang Sutawika, Eric Tang, Anish Thite, Ben Wang, Kevin Wang, and Andy Zou. The language model evaluation harness, 2024. URL <https://zenodo.org/records/12608602>.

Kimi Team, Angang Du, Bofei Gao, Bowei Xing, Changjiu Jiang, Cheng Chen, Cheng Li, and Chenjun Xiao. Kimi k1.5: Scaling reinforcement learning with LLMs. *arXiv preprint arXiv:2501.12599*, 2025. URL <https://arxiv.org/abs/2501.12599>.

Zongxi Li, Zijian Wang, Weiming Wang, Kevin Hung, Hao-ran Xie, and Fu Lee Wang. Retrieval-augmented generation for educational application: A systematic survey. *Computers and Education: Artificial Intelligence*, 8:100417, 2025. doi: 10.1016/j.caeai.2025.100417.

Niklas Muennighoff, Zitong Yang, Weijia Shi, Xiang Lisa Li, Li Fei-Fei, Hannaneh Hajishirzi, Luke Zettlemoyer, and Percy Liang. s1: Simple test-time scaling. *arXiv preprint arXiv:2501.19393*, 2025. URL <https://arxiv.org/abs/2501.19393>.

Qwen Team. Qwen2.5: A party of foundation models, September 2024. URL <https://qwenlm.github.io/blog/qwen2.5/>.

Ruohong Zhang, Tianyu Gao, Zeyu Zhu, Xinya Du, Fuli Ye, Jindong Li, Zhiyuan Chen, et al. Sky-t1: Train your own o1 preview model within \$450, 2025. URL <https://arxiv.org/abs/2501.17161>.